Aug 8 2025 BDBP106: Linux and Python programming Lab 7

Learning goals: Linux OS commands apropos, file permisisons
NOTE: Save screenshots of each exercise, and upload your work to your github account as Lab7.pdf by end of Friday Aug 8.
EXERCISES Start a fresh terminal for this Lab.

(1) The apropos command allows you to search the system's manual pages for commands that match a specific task or keyword. This is particularly useful when you need to find a command that performs a specific task but you don't know the name of the command. Try the following and display the output in your lab document.
(i) apropos "search for files"

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ apropos "search for files"
find (1)             - search for files in a directory hierarchy
suspicious-source (1) - search for files that do not meet the GPL's definitio...
ibab@IBAB-MSC-BDB-comp16-ibab:~$
```

(ii) apropos "list directory contents". Another way to get the same output is apropos "^list.*directory" (note the use of wildcards here). The ^ denotes that there is no other character before the word 'list' in the search.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ apropos "list directory contents"
dir (1)              - list directory contents
ls (1)               - list directory contents
ntfsls (8)           - list directory contents on an NTFS filesystem
vdir (1)             - list directory contents
ibab@IBAB-MSC-BDB-comp16-ibab:~$ apropos "^list.*directory"
dir (1)              - list directory contents
ls (1)               - list directory contents
ntfsls (8)           - list directory contents on an NTFS filesystem
vdir (1)             - list directory contents
ibab@IBAB-MSC-BDB-comp16-ibab:~$
```

(iii) apropos "file compress"

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ apropos "file compress"
bunzip2 (1)          - a block-sorting file compressor, v1.0.8
bzip2 (1)            - a block-sorting file compressor, v1.0.8
Dpkg::Compression::FileHandle (3perl) - class dealing transparently with file...
ibab@IBAB-MSC-BDB-comp16-ibab:~$
```

(iv) apropos "file.*copy$" (The $ denotes that there is no character to be searched after the word 'copy'.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ apropos "file.*copy$"
rcp (1)              - OpenSSH secure file copy
scp (1)              - OpenSSH secure file copy
ibab@IBAB-MSC-BDB-comp16-ibab:~$
```

Come up with 3 more keywords/phrases that you can try for this command and got meaningful output (show the screenshots of all your outputs)

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apropos “and directory”

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ apropos "and directory"
cups-files.conf (5)  - file and directory configuration file for cups
ibab@IBAB-MSC-BDB-comp16-ibab:~$
```

apropos “list directory”

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ apropos "list directory"
dir (1)              - list directory contents
ls (1)               - list directory contents
ntfsls (8)           - list directory contents on an NTFS filesystem
vdir (1)             - list directory contents
```

apropos “and files”

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ apropos "and files"
aec (1)              - compress or expand files
gunzip (1)           - compress or expand files
gzip (1)             - compress or expand files
intro (5)            - introduction to file formats and filesystems
lzop (1)             - compress or expand files
selinux_binary_policy_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
selinux_contexts_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
selinux_current_policy_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
selinux_default_context_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
selinux_default_type_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
selinux_failsafe_context_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
selinux_file_context_homedir_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
selinux_file_context_local_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
selinux_file_context_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
selinux_homedir_context_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
selinux_media_context_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
selinux_netfilter_context_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
selinux_path (3)     - These functions return the paths to the active SELinux policy configuration directories and files
selinux_removable_context_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
selinux_securetty_types_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
selinux_user_contexts_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
selinux_usersconf_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
selinux_x_context_path (3) - These functions return the paths to the active SELinux policy configuration directories and files
setup (2)            - setup devices and filesystems, mount root filesystem
swapoff (8)          - enable/disable devices and files for paging and swapping
swapon (8)           - enable/disable devices and files for paging and swapping
tee (1)              - read from standard input and write to standard output and files
uncompress (1)       - compress or expand files
zcat (1)             - compress or expand files
```

(2) File permisisons. Try the following exercises and explain the output in your work.
(i) touch newfile followed by ls -al newfile. Describe the permissions set for the user, group and everyone else.

Permission set:
user: read and write -rw -
group: read and write -rw -
everyone: read -r --

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```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ touch newfile
ibab@IBAB-MSC-BDB-comp16-ibab:~$ ls -al
total 156
drwxr-x--- 27 ibab ibab  4096 Aug  8 16:07  .
drwxr-xr-x  7 root root  4096 Jul 15 15:01  ..
-rw-r--r--  1 ibab ibab 12288 Jul 23 09:32  .23.swp
-rw-------  1 ibab ibab  2209 Aug  8 15:06  .bash_history
-rw-r--r--  1 ibab ibab   220 Jul 15 15:01  .bash_logout
-rw-r--r--  1 ibab ibab  3771 Jul 15 15:01  .bashrc
-rw-------  1 ibab ibab 12288 Aug  7 14:32  .bash.rc.swp
drwx------ 15 ibab ibab  4096 Jul 17 15:14  .cache
drwxrwxr-x  3 ibab ibab  4096 Jul 17 15:15  .chimera
drwx------ 23 ibab ibab  4096 Jul 30 09:51  .config
drwxr-xr-x  3 ibab ibab  4096 Aug  5 11:09  Desktop
drwxr-xr-x  2 ibab ibab  4096 Jul 16 16:21  Documents
drwxr-xr-x  4 ibab ibab  4096 Aug  8 14:55  Downloads
drwx------  2 ibab ibab  4096 Jul 17 15:05  .gnupg
drwxrwxr-x  2 ibab ibab  4096 Jul 17 14:59  .igv
drwxrwxr-x  3 ibab ibab  4096 Jul 17 14:59  igv
drwxrwxr-x  4 ibab ibab  4096 Jul 17 14:59  .java
drwxrwxr-x  7 ibab ibab  4096 Aug  1 14:48  JavaMorph
-rw-------  1 ibab ibab    20 Aug  8 15:55  .lesshst
drwx------  3 ibab ibab  4096 Jul 16 16:21  .local
drwxrwxr-x  3 ibab ibab  4096 Jul 17 15:06 'MEGA X'
drwxr-xr-x  2 ibab ibab  4096 Jul 16 16:21  Music
drwx------  3 ibab ibab  4096 Aug  1 14:47  .mysql
-rw-rw-r--  1 ibab ibab     0 Aug  8 16:07  newfile
```

(ii) Print the output of umask command. Explain what is the actual level of default permission based on this output for a file, and for a directory

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ umask
0002
```

0002

Actual level of default permission for a file : 0666-0002=0664 where 6 on user and group signifies read and write permission and 4 on everyone permits for read only.

Here umask is executed only for file created.

Actual level of default permission for a directory : 0777-0002=0775 where 7 on user and group signifies read write and executable permission and 5 on everyone permits for read and execute only.

(iii) Create a new file in your Lab7 directory and print and explain the current permissions for the file. Write down the octal values for the user, group and everyone

Permission set: user: read and write -rw - group: read and write -rw - everyone: read -r --

Octal value : 0664

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ mkdir lab7
ibab@IBAB-MSC-BDB-comp16-ibab:~$ cd lab7
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7$ touch labsums
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7$ ls -l
total 0
-rw-rw-r-- 1 ibab ibab 0 Aug  8 16:13 labsums
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7$
```

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(iv) Create a new directory in your Lab7 directory and print and and explain the current permissions for the directory. Write down the octal values for the user, group and everyone else.
Permission set: user: read write executable-rwx - group: read write executable -rwx - everyone: read and executable -r -x
Octal value : 0775

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7$ mkdir lab7dir
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7$ ls -l
total 4
drwxrwxr-x 2 ibab ibab 4096 Aug  8 16:16 lab7dir
-rw-rw-r-- 1 ibab ibab    0 Aug  8 16:13 labsums
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7$
```

(v) Copy the new file from above to the new directory and list the permissions of the file inside the directory. Is it the same as the file outside the directory? What does this tell you about the copying action?

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7$ cp labsums lab7dir
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7$ ls -l
total 4
drwxrwxr-x 2 ibab ibab 4096 Aug  8 16:21 lab7dir
-rw-rw-r-- 1 ibab ibab    0 Aug  8 16:13 labsums
```

Yes permissions are as same as the file outside the directory. No changes are made in the permission sets on copying action

(vi) Change the permissions of the directory such that it is only readable writable and executable by the user and the group, and only readable by everyone else. What happens to the permissions of the file inside the directory? Does it inherit the new permissions you just set? What are the new octal values for the directory permissions?

The file inside the directory doesnt have any changes on file permission. The directory has inherited the new permissions that i have just set. The new octal value for the directory permission: 0774; readable writable and executable by the user and the group, and only readable by everyone else.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7$ chmod 774 lab7dir
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7$ ls -al
total 12
drwxrwxr-x  3 ibab ibab 4096 Aug  8 16:16 .
drwxr-x--- 27 ibab ibab 4096 Aug  8 16:23 ..
drwxrwxr--  2 ibab ibab 4096 Aug  8 16:21 lab7dir
-rw-rw-r--  1 ibab ibab    0 Aug  8 16:13 labsums
```

(vii) Move the above-created new file to the new directory under a different name. Does the moving action change the file permissions?
Moving action doesnt change the file permission

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7$ mkdir lab7ethuku
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7$ mo labsums lab7ethuku
mo: command not found
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7$ mv labsums lab7ethuku
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7$ ls -al
total 16
drwxrwxr-x  4 ibab ibab 4096 Aug  8 16:40 .
drwxr-x--- 27 ibab ibab 4096 Aug  8 16:23 ..
drwxrwxr--  2 ibab ibab 4096 Aug  8 16:21 lab7dir
drwxrwxr-x  2 ibab ibab 4096 Aug  8 16:40 lab7ethuku
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7$ cd lab7ethuku
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7/lab7ethuku$ ls -al
total 8
drwxrwxr-x 2 ibab ibab 4096 Aug  8 16:40 .
drwxrwxr-x 4 ibab ibab 4096 Aug  8 16:40 ..
-rw-rw-r-- 1 ibab ibab    0 Aug  8 16:13 labsums
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab7/lab7ethuku$
```